METRIC
MIL-T-24735/1(NAVY)
31 August 1989

MILITARY SPECIFICATION SHEET

TRANSMITTER, ANALOG, FIBER OPTIC, SHIPBOARD, 0.5 TO 60 MHZ (0.5 DB PASSBAND) (METRIC)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-24735.

CLASSIFICATION

Type: 1 (LED).
Wavelength class: A (1.31 μm).
Frequency range: 2 (20 to 500 MHz).
Style: A (Dual inline package using multimode pigtail).

REQUIREMENTS

Reliability:

MTBF of 10^5 hours, where transmitter failure is defined as that time when the maximum optical power level is reduced 3 dB from the initial level, when measured at the same temperature.

Design and Construction:

Cable pigtail:

The cable pigtail used in the construction of the transmitter shall be in accordance with MIL-C-85045 and MIL-C-85045/14, using a 62.5/125 μm fiber surrounded by aramid strength members and an outer jacket.

AMSC N/A
FSC 6025
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Fiber optic connector: N/A.

Case:

Transmitter dimensions shall be as shown on figure 1. The pinouts of the hermetically sealed transmitter shall be as shown in table I.

Terminals:

Terminal strength:

Failure criteria shall be as specified in MIL-STD-883, method 2004. Electrical terminals shall be subjected to tension (A), bending stress (B1) and lead fatigue (B2) tests.

Performance:

Peak emission wavelength: 1.27 to 1.35 µm.
Spectral width: 0.16 µm (maximum).

Optical power levels:

Maximum: Minus 9 decibels referred to 1 milliwatt (dBmW) at 25°C out of the specified 62.5/125 µm fiber with maximum input signal.
Minimum: Minus 12 dBmW at 25°C out of the specified 62.5/125 µm fiber with maximum input signal.

Electro-optical characteristics:

Signal-to-noise ratio (SNR): 14 dB minimum.
Frequency range: 0.5 MHz to 60 MHz
Total harmonic distortion (THD): Less than 2 percent.
Intermodulation distortion (IMD): Greater than 50 dB below the specified input signal.
Amplitude distortion: Plus or minus 0.5 dB deviation over frequency range.
Group delay: 0.3 ns (maximum).

Analog electrical input characteristics:

Voltage: 2 volt peak to peak.
Waveform: Sinusoidal.
Impedance: 50 ohms plus or minus 1 percent.

Power supply voltage and current: Plus 12.0 ± 1.5 volts, 200 milliamperes (mA) (maximum).

Environmental:

Temperature extremes: Range 2 (operating: minus 28 to plus 85°C, nonoperating: minus 62 to plus 85°C).

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Mechanical:

Connector pull-out force: N/A.
Fiber dynamic strength: N/A.

Electromagnetic effects: Below-deck application, in accordance with MIL-STD-461, part 5.

Conducted susceptibility: CS01, CS02, CS06, CS09.
Conducted emission: CE01, CE03, CE06.
Radiated emission: RE01, RE02, RE03.
Radiated susceptibility: RS01, RS02, RS03.

Dimensions: See figure 1.

Identification marking:

The part number and part designation as specified in 6.8 and 6.9 of MIL-T-24735 shall be marked on the transmitter case, along with the following: inspection lot identification code, manufacturer's identification mark/designating symbol, pin 1 identifier, and electrostatic identifier (if required), all as defined in MIL-M-38510.

QUALITY ASSURANCE PROVISIONS

Screening:

All transmitters to be delivered or submitted for first article inspection shall have been subjected to and passed all of the specified screening tests. Monitoring of electrical/optical parameters are not required during the screening tests; however, after completion of all screening tests, the transmitters shall be subjected to group I or group A testing.

Burn-in: Test condition B, with an ambient burn-in test temperature of plus 85°C for a minimum of 160 hours. The power supply voltage shall be not less than plus 12.5 volts and the transmitter shall be operated at the maximum optical power level.

Temperature cycling: In step 1, use minus 62°C (plus 0, minus 10) and in step 2 use plus 85°C (plus 10, minus 0), for 10 cycles.

Reliability: Test temperature shall be plus 85°C (minimum).

Design and construction inspections:

Terminal strength:

Tension (test condition A): 5 newtons (minimum) for a period of 5 minutes (minimum).
Bending stress (test condition B1): Use procedure for DIP leads.
Lead fatigue (test condition B2):

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Use procedure for DIP leads with force sufficient to bend the lead as specified in B1.

Solderability:

Three terminals on each transmitter shall be examined in accordance with MIL-STD-883, method 2003, with a soldering temperature of 245 plus or minus 5°C.

Performance inspections:

Optical performance tests

Optical output properties:

For step 5, use the maximum analog input magnitude specified.

Electro-optical characteristics:

For step 4, use a sinusoidal input at the frequencies specified for each parameter.

Step 5 passband filter: N/A.

The SNR shall be verified over the specified operating frequency range at frequencies starting with 0.5 MHz. Additional SNR testing shall be at the following frequencies: 1.0 MHz, 2.0 MHz, 4.0 MHz, 8.0 MHz, 15.0 MHz, 30.0 MHz, and 60.0 MHz.

For the THD test, the fundamental input frequencies shall be in the following order: f = 0.5 MHz; f = 1.0 MHz; f = 2.0 MHz; f = 4.0 MHz; f = 8.0 MHz; f = 15.0; f = 30 MHz; and f = 60 MHz. At each input frequency, m = 5. The THD measured at f = 30 MHz shall be less than the THD at f = 15 MHz. The THD measured at f = 60 MHz shall be less than the THD at f = 30 MHz.

For the IMD test, the test frequencies shall be in accordance with the following: Initial value of f_1 = 0.5 MHz; Initial value of f_2 = 1.0 Mhz. $f_{1\ next} = f_1 + 0.5$ MHz; $f_{2\ next} = f_2 + 0.5$ MHz. f_2 shall not exceed 60 MHz.

Group delay test: Verify requirements at 1 MHz steps over the frequency range specified.

Power supply voltage and current:

The current shall be measured at standard test conditions at the minimum power supply voltage.

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Environmental inspections:

Explosive atmosphere: The values for the test conditions shall be

Fuel - n-hexane (reagent grade or 95 percent).
Fuel amount - 3.8 percent by volume of the test chamber.
Test altitude - Sea level.
Test temperature - Plus 85°C.

Nuclear radiation resistance:

Total dose: Three transmitters shall be subjected to this test, each tested at different temperatures: minus 28, plus 20 and plus 85°C. The total dose shall be provided by the contract activity.

Neutron: One transmitter shall be subjected to this test at plus 20°C.

Electromagnetic effects:

Tests shall be conducted in accordance with MIL-STD-462, over a frequency range of 14 to 500 kHz. The RS03 limit shall be 200 V/m. Tests shall be conducted at ambient conditions. Pre or post-test evaluation shall include structural integrity and spectral distribution. Optical power and SNR shall be monitored during the test and shall meet the requirements specified herein. Use the specified analog input at the maximum frequency specified.

INTENDED USE

The transmitter specified herein is intended for use in Navy land-based or shipboard applications where bandwidths are not greater than 60 MHz (single sideband composite sinusoidal waveforms) are required. These transmitters are intended to be compatible with receivers specified in MIL-R-24737 and MIL-R-24737/1.

PART NUMBER: D24735/01-001.

PART DESIGNATOR: 1A2A.

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6025-N002-01)

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TABLE I. Pin assignment.

Pin number	Function
1	ground
2	ground
3	data
4	1/
5	ground
6	ground
7	ground
8	ground
9	ground
10	1/
11	1/
12	supply voltage
13	1/
14	supply voltage
15	ground
16	ground

1/ Internal connection reserved for manufacturer.

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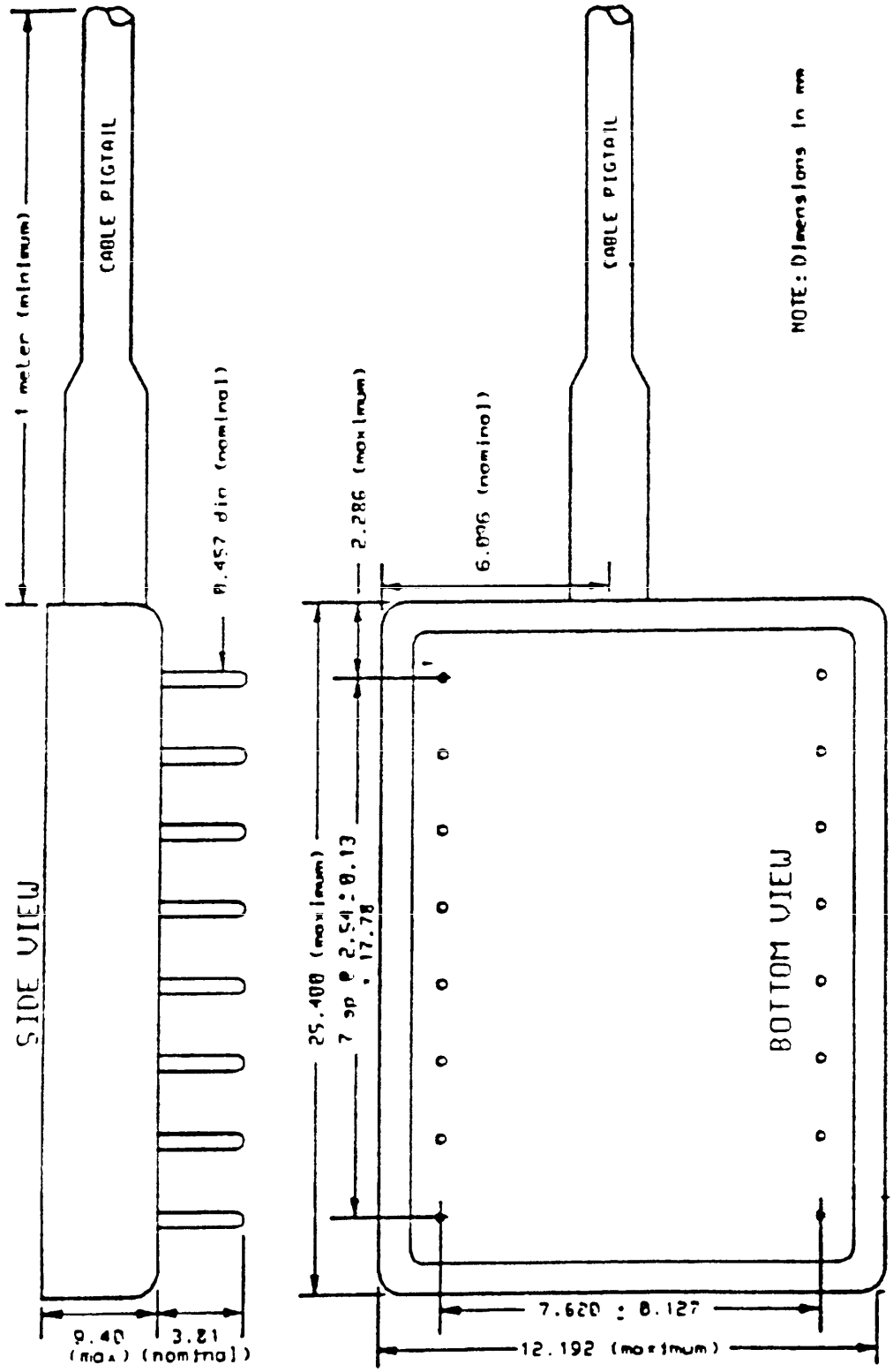


FIGURE 1. Transmitter case outline; 16 pin DIP with multimode pigtail.

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